TASK 1 : Understanding and using basic HDFS commands

Apache Hadoop hadoop fs or hdfs dfs are file system commands to interact with HDFS, these commands are very similar to Unix Commands. Note that some Syntax and output formats may differ between Unix and HDFS Commands.
Hadoop is a open-source distributed framework that is used to store and process a large set of datasets. To store data, Hadoop uses HDFS, and to process data, it uses MapReduce & Yarn. In this article, I will mainly focus on Hadoop HDFS commands to interact with the files.

Hadoop provides two types of commands to interact with File System; hadoop fs or hdfs dfs. Major difference being hadoop commands are supported with multiple file systems like S3, Azure and many more.

Start Hadoop Services
In order to run hdfs dfs or hadoop fs commands, first, you need to start the Hadoop services by running the start-dfs.sh script from the Hadoop installation.

ubuntu@namenode:~$ start-dfs.sh
Starting namenodes on [namenode.socal.rr.com]
Starting datanodes
Starting secondary namenodes [namenode]
ubuntu@namenode:~$

Basic HDFS DFS Commands

Below are basic hdfs dfs or hadoop fs Commands.

COMMAND	DESCRIPTION
-ls	List files with permissions and other details
-mkdir	Creates a directory named path in HDFS
-rm	To Remove File or a Directory
-rmr	Removes the file that identified by path / Folder and subfolders
-rmdir	Delete a directory
-put	Upload a file / Folder from the local disk to HDFS
-cat	Display the contents for a file
-du	Shows the size of the file on hdfs.
-dus	Directory/file of total size
-get	Store file / Folder from HDFS to local file
-getmerge	Merge Multiple Files in an HDFS

-count	Count number of directory, number of files and file size
-setrep	Changes the replication factor of a file
-mv	HDFS Command to move files from source to destination
-moveFromLocal	Move file / Folder from local disk to HDFS
-moveToLocal	Move a File to HDFS from Local
-cp	Copy files from source to destination
-tail	Displays last kilobyte of the file
-touch	create, change and modify timestamps of a file
-touchz	Create a new file on HDFS with size 0 bytes
-appendToFile	Appends the content to the file which is present on HDF
-copyFromLocal	Copy file from local file system
-copyToLocal	Copy files from HDFS to local file system
-usage	Return the Help for Individual Command
-checksum	Returns the checksum information of a file
-chgrp	Change group association of files/change the group of a file or a path
-chmod	Change the permissions of a file
-chown	change the owner and group of a file
-df	Displays free space
-head	Displays first kilobyte of the file
-Create Snapsh ots	Create a snapshot of a snapshottable directory
-Delete Snapsh ots	Delete a snapshot of from a snapshottable directory
-Rename Snapsh ots	Rename a snapshot
-expunge	create new checkpoint
-Stat	Print statistics about the file/directory
-truncate	Truncate all files that match the specified file pattern to the specified length
-find	Find File Size in HDFS

ls – List Files and Folder

HDFS ls command is used to display the list of Files and Directories in HDFS, This ls command shows the files with permissions, user, group, and other details. For more information follow ls- List Files and Folder

```
$hadoop fs -ls
```

```
or
$hdfs dfs -ls
```

mkdir – Make Directory

HDFS mkdir command is used to create a directory in HDFS. By default, this directory would be owned by the user who is creating it. By specifying "/" at the beginning it creates a folder at root directory.

```
$hadoop fs -mkdir /directory-name or
$hdfs dfs -mkdir /directory-name
```

rm – Remove File or Directory

HDFS rm command deletes a file and a directory from HDFS recursively.

```
$hadoop fs -rm /file-name or
$hdfs dfs -rm /file-name
```

rmr – Remove Directory Recursively

Rmr command is used to deletes a file from Directory recursively, it is a very useful command when you want to delete a non-empty directory.

```
$hadoop fs -rmr /directory-name or
$hdfs dfs -rmr /directory-name
```

rmdir – Delete a Directory

Rmdir command is used to removing directories only if they are empty.

```
$hadoop fs -rmdir /directory-name or
$hdfs dfs -rmdir /directory-name
```

put – Upload a File to HDFS from Local

Copy file/folder from local disk to HDFS. On put command specifies the local-file-path where you wanted to copy from and then where hdfs-file-path ed to copy to on hdfs.

```
$ hadoop fs -put /local-file-path /hdfs-file-path or
$ hdfs dfs -put /local-file-path /hdfs-file-path
```

cat – Displays the Content of the File

The cat command reads the specified file from HDFS and displays the content of the file on console or stdout.

```
$ hadoop fs -cat /hdfs-file-path or
$ hdfs dfs -cat /hdfs-file-path
```

du – File Occupied in Disk

Du command is used to How much file Occupied in the disk. The field is the base size of the file or directory before replication.

```
$ hadoop fs -du  /hdfs-file-path or
$ hdfs dfs -du  /hdfs-file-path
```

dus – Directory/file of the total size

Dus command is used to will give the total size of directory/file.

```
$ hadoop fs -dus  /hdfs-directory  or
$ hdfs dfs -dus  /hdfs-directory
```

get – Copy the File from HDFS to Local

Get command is used to store filess from HDFS to the local file. HDFS file gets the local machine.

```
$ hadoop fs -get /local-file-path /hdfs-file-path or
$ hdfs dfs -get /local-file-path /hdfs-file-path
```

getmerge – Merge Multiple Files in an HDFS

If you have multiple files in an HDFS, use -getmerge option command. All these multiple files merged into one single file and downloads to local file system.

```
$ hadoop fs -getmerge [-nl] /source /local-destination or
$ hdfs dfs -getmerge [-nl] /source /local-destination
```

count – Number of Directory

The count command is used to count a number of directories, a number of files, and file size on HDFS.

```
$ hadoop fs -count /hdfs-file-path or
$ hdfs dfs -count /hdfs-file-path
```

mv – Moves Files from Source to Destination

MV (move) command is used to move files from one location to another location in HDFS. Move command allows multiple sources as well in which case the destination needs to be a director.

```
$ hadoop fs -mv /local-file-path /hdfs-file-path or
$ hdfs dfs -mv /local-file-path /hdfs-file-path
```

moveFromLocal – Move file / Folder from Local disk to HDFS

Similar to the put command, `moveFromLocal` moves the file or source from the local file path to the destination in the HDFS file path. After this command, you will not find the file on the local file system.

```
$ hadoop fs -moveFromLocal /local-file-path /hdfs-file-path or
$ hdfs dfs -moveFromLocal /local-file-path /hdfs-file-path
```

moveToLocal – Move a File to HDFS from Local

Similar to the get command, moveToLocal moves the file or source from the HDFS file path to the destination in the local file path.

```
$ hadoop fs -moveToLocal /hdfs-file-path /local-file-path  or
$ hdfs dfs -moveToLocal /hdfs-file-path /local-file-path
```

Cp – Copy Files from Source to Destination

Copy File-one location to another location in HDFS. Copy files from source to destination, Copy command allows multiple sources as well in which case the destination must be a directory.

```
$ hadoop fs -cp /local-file-path /hdfs-file-path or
$ hdfs dfs -cp /local-file-path /hdfs-file-path
```

setrep – Changes the Replication Factor of a File

This HDFS command is used to change the replication factor of a file. If the path is a directory then the command recursively changes the replication factor of all files under the directory tree rooted at the path.

```
$ hadoop fs -setrep /number /file-name  or
$ hdfs dfs -setrep /number /file-name
```

tail – Displays Last Kilobyte of the File

Tail command is used to Display last kilobyte of the file to stdout.

```
$ hadoop fs -tail /hdfs-file-path or
$ hdfs dfs -tail /hdfs-file-path
```

touch – Create and Modify Timestamps of a File

It is used to create a file without any content. The file created using the touch command is empty. updates the access and modification times of the file specified by the URI to the current time, the file does not exist then a zerolength file is created at URI with the current time as the timestamp of that URI.

```
$ hadoop fs -touch /hdfs-file-path or
```

```
$ hdfs dfs -touch /hdfs-file-path
```

touchz – Create a File of zero Length

Create a new file on HDFS with size 0 bytes. create a file of zero length, an error is returned if the file exists with non-zero length.

```
$ hadoop fs -touchz /hdfs-file-path or
$ hdfs dfs -touchz /hdfs-file-path
```

appendToFile – Appends the Content to the File

Appends the content to the file which is present on HDFS. Append single source. or multiple sources from the local file system to the destination file system. this command appends the contents of all the given local files to the provided destination file on the HDFS filesystem.

```
$ hadoop fs -appendToFile /hdfs-file-path or
$ hdfs dfs -appendToFile /hdfs-file-path
```

copyFromLocal – Copy File from Local file System
Copying file from a local file to HDFS file system. Similar to the fs put command and copyFromLocal command both are Store files from local disk to HDFS. Except that the source is restricted to a local file reference.

```
$ hadoop fs -copyToLocal /hdfs-file-path /local-file-path or
$ hdfs dfs -copyToLocal  /hdfs-file-path /local-file-path
```

copyToLocal – Copy Files from HDFS to Local file System
Copying files from HDFS file to local file system. Similar to the fs get command and copyToLocal command both are Store files from hdfs to local files. Except that the destination is restricted to a local file reference.

```
$ hadoop fs -copyToLocal /hdfs-file-path /local-file-path or
$ hdfs dfs -copyToLocal  /hdfs-file-path /local-file-path
```

usage – Return the Help for Individual Command
Usage command is used to Provide you help for indidual commands.

```
$ hadoop fs -usage mkdir  or
$ hdfs dfs -usage mkdir
```

checksum -Returns the Checksum Information of a File
The checksum command is used to Returns the Checksum Information of a File. Returns the checksum information of a file.

```
$ hadoop fs -checksum [-v] URI or
$ hdfs dfs -checksum [-v] URI
```

chgrp – Change Group Association of Files chgrg command is used to change the group of a file or a path. The user must be the owner of files, or else a super-user.

```
$ hadoop fs -chgrp [-R] groupname or
$ hdfs dfs -chgrp [-R] groupname
```

chmod – Change the Permissions of a File
This command is used to change the permissions of a file. With -R Used to modify the files recursively and it is the only option that is being supported currently.

```
$ hadoop fs -chmod [-R] hdfs-file-path  or
$ hdfs dfs -chmod [-R] hdfs-file-path
```

chown – Change the Owner and Group of a File

Chown command is used to change the owner and group of a file. This command is similar to the shell‟s chown command with a few exceptions.

```
$ hadoop fs -chown [-R] [owner][:[group]] hdfs-file-path or
```

```
$ hdfs dfs -chown [-R] [owner][:[group]] hdfs-file-path
```

df – Displays free Space

Df is the Displays free space. This command is used to show the capacity, free and used space available on the HDFS filesystem. Used to format the sizes of the files in a human-readable manner rather than the number of bytes.

```
$ hadoop fs -df /user/hadoop/dir1 or
$ hdfs dfs -df /user/hadoop/dir1
```

head – Displays first Kilobyte of the File

Head command is use to Displays first kilobyte of the file to stdout.

```
$ hadoop fs -head /hdfs-file-path or
$ hdfs dfs -head /hdfs-file-path
```

createSnapshots – Create Snapshottable Directory

This operation requires owner privilege of the snapshot table directory. The path of the snapshot table directory, snapshot name is The snapshot name a default name is generated using a timestamp.

```
$ hadoop fs -createSnapshot /path /snapshotName or
$ hdfs dfs -createSnapshot /path /snapshotName
```

deleteSnapshots – Delect Snapshottable Directory

This operation requires owner privilege of the snapshot table directory. The path of the snapshot table directory, snapshot name is The snapshot name.

```
$ hadoop fs -deleteSnapshot /path /snapshotName or
$ hdfs dfs -deleteSnapshot /path /snapshotName
```

renameSnapshots – Rename a Snapshot

This operation requires owner privilege of the snapshottable directory.

```
$ hadoop fs -renameSnapshot /path /oldName /newName
or
$ hdfs dfs  -renameSnapshot /path /oldName /newName
```

expunge – Create New Checkpoint

This command is used to empty the trash available in an HDFS system.
Permanently delete files in checkpoints older than the retention threshold from the trash directory.

```
$ hadoop fs –expunge -immediate -fs  /hdfs-file-path or
$ hdfs dfs –expunge -immediate -fs  /hdfs-file-path
```

Stat – File/Directory Print Statistics

This command is used to print the statistics about the file/directory in the specified format. Print statistics about the file/directory at in the specified format.

```
$ hadoop fs -stat /format or
$ hdfs dfs -stat /format
```

Truncate – Specified File Pattern and Length

Truncate all files that match the specified file pattern to the specified length.

```
$ hadoop fs -truncate [-w] /length /hdfs-file-path or
$ hdfs dfs -truncate [-w] /length /hdfs-file-path
```

Find – Find File Size in HDFS

In Hadoop, `hdfs dfs -find` or `hadoop fs -find` commands are used to get the size of a single file or size for all files specified in an expression or in a directory.
By default, it points to the current directory when the path is not specified.

```
$hadoop fs -find / -name test -print or
$hdfs dfs -find / -name test -print
```